

Table S4. Deep molecular-model settings. All models used batches of 32. GNN denotes the four pretrained architectures: GCN, GAT, GIN, and GraphSAGE. These frozen molecular-model settings, including their original class weighting, are unchanged; unweighted BCE applies to the revised transcriptomic stage. Shared settings: training seed: 1.

Representation	Size	Model	Selection / tune	Class weighting	Settings
Molecular graph	300.0	GCN	Initialized from supervised context; checkpoint; end-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	5 message-passing layers; hidden_dim=300; pooling; JK=last; dropout=0.5; batch_size=32; Adam; lr=1e-4; max_epochs=100; patience=20
Molecular graph	300.0	GAT	Initialized from supervised context; checkpoint; end-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	5 message-passing layers; hidden_dim=300; pooling; JK=last; dropout=0.5; batch_size=32; Adam; lr=1e-4; max_epochs=100; patience=20
Molecular graph	300.0	GIN	Initialized from supervised context; checkpoint; end-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	5 message-passing layers; hidden_dim=300; pooling; JK=last; dropout=0.5; batch_size=32; Adam; lr=1e-4; max_epochs=100; patience=20
Molecular graph	300.0	GraphSAGE	Initialized from supervised context; checkpoint; end-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	5 message-passing layers; hidden_dim=300; pooling; JK=last; dropout=0.5; batch_size=32; Adam; lr=1e-4; max_epochs=100; patience=20
Canonical SMILES tokens		DeepChem/Chem	End-to-end single-task fine-tuning; validation AUPRC checkpoint select	BCE pos_weight=n_neg	max_length=128; batch_size=32; Adam; lr=1e-4; weight_decay=0; linear warmup=6; gradient clipping=1.0; max_epochs=20; patience=5
Directed molecular graph	300.0	Chemprop D-MPN	Single-task training; early stopping on validation loss	Unweighted binary classification loss	hidden_dim=300; depth=3; mean aggregation; dropout=0; batch_size=32; max_epochs=100; patience=20